

Communication entre la VM Linux Docker et la VM pfSense :

```
Apr 12 18:12:25 SRV-Docker systemd[1]: Started docker.service - Docker Ap
root@SRV-Docker:~# ping -c 4 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=2.37 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=2.09 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=2.25 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=1.99 ms

--- 192.168.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.985/2.175/2.373/0.149 ms
root@SRV-Docker:~# █
```

Communication avec le WAN (Accès Internet) :

```
--- 192.168.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.985/2.175/2.373/0.149 ms
root@SRV-Docker:~# ping -c 4 google.com
PING google.com (142.251.39.110) 56(84) bytes of data.
64 bytes from pnpara-ag-in-f14.1e100.net (142.251.39.110): icmp_seq=1 ttl=116 ti
me=8.87 ms
64 bytes from pnpara-ag-in-f14.1e100.net (142.251.39.110): icmp_seq=2 ttl=116 ti
me=8.15 ms
64 bytes from pnpara-ag-in-f14.1e100.net (142.251.39.110): icmp_seq=3 ttl=116 ti
me=9.67 ms
64 bytes from pnpara-ag-in-f14.1e100.net (142.251.39.110): icmp_seq=4 ttl=116 ti
me=8.02 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 7460ms
rtt min/avg/max/mdev = 8.024/8.678/9.672/0.657 ms
root@SRV-Docker:~# █
```

Fonctionnement de la Résolution DNS en affichant l'IP de Google :

```
root@SRV-Docker:~# nslookup google.com
Server:      192.168.1.1
Address:     192.168.1.1#53
```

Non-authoritative answer:

```
Name:   google.com
Address: 142.251.39.110
Name:   google.com
Address: 2a00:1450:4007:810::200e
```

Ce fichier orchestre les trois conteneurs sur le réseau interne **Webnet**.

Configuration du fichier Docker-Compose (docker-compose.yaml) :

```
services:
  nginx1:
    image: nginx
    container_name: nginx1
    networks:
      - webnet
    ports:
      - "8081:80"
    volumes:
      - ./nginx1.conf:/etc/nginx/nginx.conf
  nginx2:
    image: nginx
    container_name: nginx2
    networks:
      - webnet
    ports:
      - "8082:80"
    volumes:
      - ./nginx2.conf:/etc/nginx/nginx.conf

loadbalancer:
  image: haproxy:alpine
  container_name: loadbalancer
  networks:
    - webnet
  ports:
    - "8080:80"
  volumes:
    - ./haproxy.cfg:/usr/local/etc/haproxy/haproxy.cfg

networks:
  webnet:
    driver: bridge
```

Configuration du répartiteur de charge avec l'algorithme **Round-robin** pour alterner entre les serveurs.

Configuration du fichier HAProxy (nano haproxy.cfg) :

```
defaults
  mode http
  timeout connect 5000ms
  timeout client 50000ms
  timeout server 50000ms

frontend http_front
  bind *:80
  default_backend http_back

backend http_back
  balance roundrobin
  server nginx1 nginx1:80 check
  server nginx2 nginx2:80 check
```

Configurations des serveurs Nginx complètes (nano nginx1.conf & nginx2.conf) :

```
events {}
http {
  server {
    listen 80;
    location / {
      default_type text/plain;
      add_header Cache-control "no-store";
      return 200 'Hello from nginx1';
    }
  }
}
```

```
events {}
http {
    server {
        listen 80;
        location / {
            default_type text/plain;
            add_header Cache-control "no-store";
            return 200 'Hello from nginx2';
        }
    }
}
```

Lancement de la Stack et vérification des conteneurs :

```
potentia
[+] Running 16/16
  ✓ loadbalancer Pulled 66.5s
  ✓ 589002ba0eae Pull complete 40.1s
  ✓ 2c881f47edac Pull complete 39.0s
  ✓ 1a99bb61f6ef Pull complete 38.1s
  ✓ e562e26d97bf Pull complete 50.0s
  ✓ b681ddf788cd Pull complete 40.0s
  ✓ 4f4fb700ef54 Pull complete 41.3s
  ✓ nginx1 Pulled 66.8s
  ✓ nginx2 Pulled 66.9s
  ✓ 5435b2dcd5fc Pull complete 32.2s
  ✓ 054715a6bffa Pull complete 34.9s
  ✓ 88d1d984b765 Pull complete 4.9s
  ✓ 4a038fd18db1 Pull complete 25.2s
  ✓ 84e114c2bb36 Pull complete 33.6s
  ✓ 7b5d674621c2 Pull complete 35.6s
  ✓ 448ea5cac5d5 Pull complete 35.6s
[+] Running 4/4
  ✓ Network stack-web_webnet Created 1.5s
  ✓ Container loadbalancer Started 1.0s
  ✓ Container nginx1 Started 1.1s
  ✓ Container nginx2 Started 1.1s
root@SRV-Docker:~/stack-web# sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
a473a01e7587   haproxy:alpine "docker-entrypoint.s..." 25 seconds ago Up 18 seconds 0.0.0.0:8080->80/tcp, :::8080->80/tcp
loadbalancer
0fb0d60f2b7e   nginx         "/docker-entrypoint...." 25 seconds ago Up 18 seconds 0.0.0.0:8081->80/tcp, :::8081->80/tcp
```

Test d'alternance entre nginx1 et nginx2 :

```
root@SRV-Docker:~/stack-web# sudo docker-compose up -d --force-recreate
WARN[0000] /root/stack-web/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potentia
l confusion
[+] Running 3/3
  ✓ Container loadbalancer Started 3.9s
  ✓ Container nginx1 Started 2.8s
  ✓ Container nginx2 Started 0.8s
root@SRV-Docker:~/stack-web# curl http://localhost:8080
Hello from nginx1root@SRV-Docker:~/stack-web# curl http://localhost:8082
Hello from nginx2root@SRV-Docker:~/stack-web# curl http://localhost:8081
```

Test de charge : Utilisation d'Apache Bench (ab -n 1000 -c 10) pour valider la stabilité du cluster sous charge :

```
root@SRV-Docker:~/stack-web# ab -n 1000 -c 10 http://localhost:8080/
This is ApacheBench, Version 2.3 <Revision: 1923142 >
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking localhost (be patient)
Completed 100 requests
Completed 200 requests
Completed 300 requests
Completed 400 requests
Completed 500 requests
Completed 600 requests
Completed 700 requests
Completed 800 requests
Completed 900 requests
Completed 1000 requests
Finished 1000 requests

Server Software:      nginx/1.29.8
Server Hostname:      localhost
Server Port:          8080

Document Path:        /
Document Length:      17 bytes

Concurrency Level:    10
Time taken for tests:  1.428 seconds
Complete requests:    1000
```

```

Failed requests:      0
Total transferred:    185000 bytes
HTML transferred:     17000 bytes
Requests per second:  700.24 [#/sec] (mean)
Time per request:     14.281 [ms] (mean)
Time per request:     1.428 [ms] (mean, across all concurrent requests)
Transfer rate:        126.51 [Kbytes/sec] received

Connection Times (ms)
      min    mean[+/-sd] median    max
Connect:    0       1   1.5      0     11
Processing:  1      13   7.9     12     77
Waiting:    0       13   7.2     11     69
Total:      3       14   8.1     12     77

Percentage of the requests served within a certain time (ms)
 50%    12
 66%    15
 75%    17
 80%    19
 90%    24
 95%    27
 98%    38
 99%    41
100%    77 (longest request)

```

Test de résilience : Arrêt forcé de nginx1. HAProxy a immédiatement redirigé 100% du trafic vers nginx2 sans interruption pour l'utilisateur :

```

root@SRV-Docker:~/stack-web# docker stop nginx1
nginx1
root@SRV-Docker:~/stack-web# curl http://localhost:8080
Hello from nginx2root@SRV-Docker:~/stack-web# docker start nginx1
nginx1
root@SRV-Docker:~/stack-web# curl http://localhost:8080
Hello from nginx2root@SRV-Docker:~/stack-web# curl http://localhost:8080:8080/
Hello from nginx2root@SRV-Docker:~/stack-web# curl http://localhost:8081

```

Centralisation et automatisation des logs :

```

root@SRV-Docker:~/stack-web# sudo mkdir -p /var/logs/backups
root@SRV-Docker:~/stack-web# sudo chmod 750 /var/logs/backups
root@SRV-Docker:~/stack-web# ls -lh /var/logs/backups
total 0
root@SRV-Docker:~/stack-web# sudo ls -lh /var/logs/backups
total 0

Sauvegarde terminée le 2026-04-13_05-23
root@SRV-Docker:~/stack-web# ls -l /var/logs/backups
total 0
root@SRV-Docker:~/stack-web# crontab -e
crontab: installing new crontab
root@SRV-Docker:~/stack-web# crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
* * * * * /bin/bash /root/stack-web/save-logs.sh

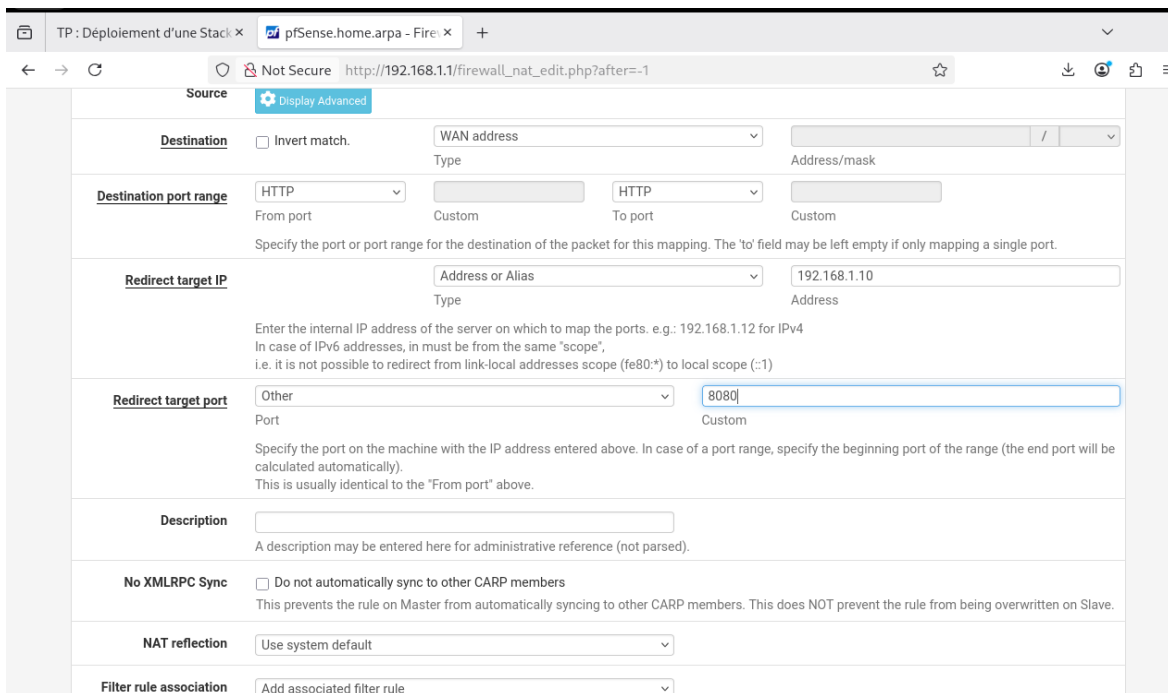
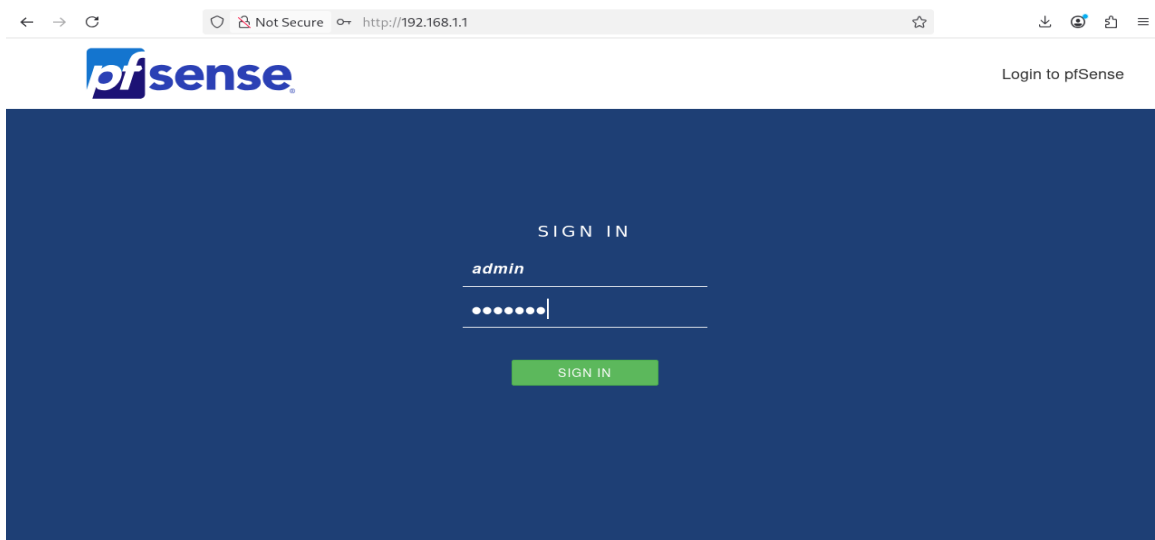
```

Automatisation de la Journalisation avec les sauvegardes horodatées s'exécutant chaque minute via la tâche Cron :

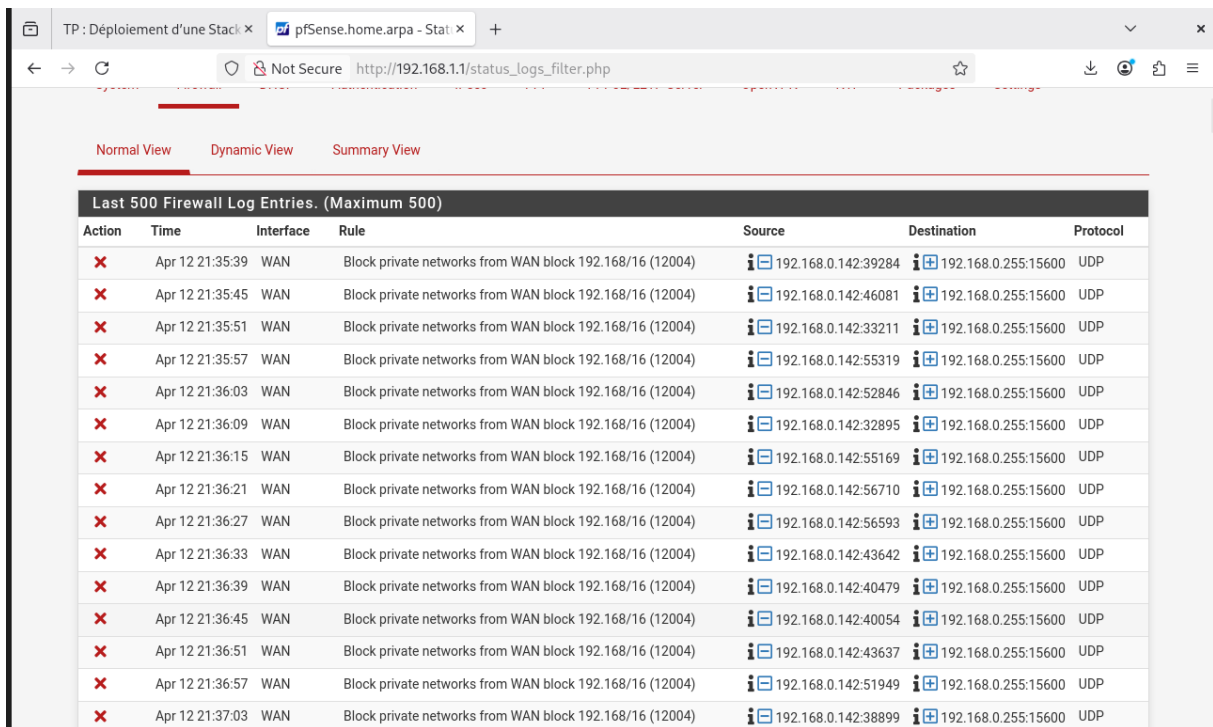
```
drwxr-x--- 2 root root 4096 Apr 13 06:10 /var/logs/backups
root@SRV-Docker:~/stack-web# ls -l /var/logs/backups
total 1056
-rw-r--r-- 1 root root      0 Apr 13 06:01 haproxy_2026-04-13_06-01.log
-rw-r--r-- 1 root root      0 Apr 13 06:02 haproxy_2026-04-13_06-02.log
-rw-r--r-- 1 root root      0 Apr 13 06:03 haproxy_2026-04-13_06-03.log
-rw-r--r-- 1 root root      0 Apr 13 06:04 haproxy_2026-04-13_06-04.log
-rw-r--r-- 1 root root      0 Apr 13 06:05 haproxy_2026-04-13_06-05.log
-rw-r--r-- 1 root root      0 Apr 13 06:06 haproxy_2026-04-13_06-06.log
-rw-r--r-- 1 root root      0 Apr 13 06:07 haproxy_2026-04-13_06-07.log
-rw-r--r-- 1 root root      0 Apr 13 06:08 haproxy_2026-04-13_06-08.log
-rw-r--r-- 1 root root      0 Apr 13 06:09 haproxy_2026-04-13_06-09.log
-rw-r--r-- 1 root root      0 Apr 13 06:10 haproxy_2026-04-13_06-10.log
-rw-r--r-- 1 root root      0 Apr 13 06:11 haproxy_2026-04-13_06-11.log
-rw-r--r-- 1 root root 46645 Apr 13 06:01 nginx1_2026-04-13_06-01.log
-rw-r--r-- 1 root root 46645 Apr 13 06:02 nginx1_2026-04-13_06-02.log
-rw-r--r-- 1 root root 46645 Apr 13 06:03 nginx1_2026-04-13_06-03.log
-rw-r--r-- 1 root root 46645 Apr 13 06:04 nginx1_2026-04-13_06-04.log
-rw-r--r-- 1 root root 46645 Apr 13 06:05 nginx1_2026-04-13_06-05.log
-rw-r--r-- 1 root root 46645 Apr 13 06:06 nginx1_2026-04-13_06-06.log
-rw-r--r-- 1 root root 46645 Apr 13 06:07 nginx1_2026-04-13_06-07.log
-rw-r--r-- 1 root root 46645 Apr 13 06:08 nginx1_2026-04-13_06-08.log
-rw-r--r-- 1 root root 46645 Apr 13 06:09 nginx1_2026-04-13_06-09.log
-rw-r--r-- 1 root root 46645 Apr 13 06:10 nginx1_2026-04-13_06-10.log
-rw-r--r-- 1 root root 46645 Apr 13 06:11 nginx1_2026-04-13_06-11.log
-rw-r--r-- 1 root root 46099 Apr 13 06:01 nginx2_2026-04-13_06-01.log
-rw-r--r-- 1 root root 46099 Apr 13 06:02 nginx2_2026-04-13_06-02.log
-rw-r--r-- 1 root root 46099 Apr 13 06:03 nginx2_2026-04-13_06-03.log
-rw-r--r-- 1 root root 46099 Apr 13 06:04 nginx2_2026-04-13_06-04.log
-rw-r--r-- 1 root root 46099 Apr 13 06:05 nginx2_2026-04-13_06-05.log
-rw-r--r-- 1 root root 46099 Apr 13 06:06 nginx2_2026-04-13_06-06.log
-rw-r--r-- 1 root root 46099 Apr 13 06:07 nginx2_2026-04-13_06-07.log
-rw-r--r-- 1 root root 46099 Apr 13 06:08 nginx2_2026-04-13_06-08.log
```

```
root@SRV-Docker:~/stack-web# ls -lh /var/logs/backups
total 1.3M
-rw-r--r-- 1 root root      0 Apr 13 06:01 haproxy_2026-04-13_06-01.log
-rw-r--r-- 1 root root      0 Apr 13 06:02 haproxy_2026-04-13_06-02.log
-rw-r--r-- 1 root root      0 Apr 13 06:03 haproxy_2026-04-13_06-03.log
-rw-r--r-- 1 root root      0 Apr 13 06:04 haproxy_2026-04-13_06-04.log
-rw-r--r-- 1 root root      0 Apr 13 06:05 haproxy_2026-04-13_06-05.log
-rw-r--r-- 1 root root      0 Apr 13 06:06 haproxy_2026-04-13_06-06.log
-rw-r--r-- 1 root root      0 Apr 13 06:07 haproxy_2026-04-13_06-07.log
-rw-r--r-- 1 root root      0 Apr 13 06:08 haproxy_2026-04-13_06-08.log
-rw-r--r-- 1 root root      0 Apr 13 06:09 haproxy_2026-04-13_06-09.log
-rw-r--r-- 1 root root      0 Apr 13 06:10 haproxy_2026-04-13_06-10.log
-rw-r--r-- 1 root root      0 Apr 13 06:11 haproxy_2026-04-13_06-11.log
-rw-r--r-- 1 root root      0 Apr 13 06:12 haproxy_2026-04-13_06-12.log
-rw-r--r-- 1 root root      0 Apr 13 06:13 haproxy_2026-04-13_06-13.log
-rw-r--r-- 1 root root 46K Apr 13 06:01 nginx1_2026-04-13_06-01.log
-rw-r--r-- 1 root root 46K Apr 13 06:02 nginx1_2026-04-13_06-02.log
-rw-r--r-- 1 root root 46K Apr 13 06:03 nginx1_2026-04-13_06-03.log
-rw-r--r-- 1 root root 46K Apr 13 06:04 nginx1_2026-04-13_06-04.log
-rw-r--r-- 1 root root 46K Apr 13 06:05 nginx1_2026-04-13_06-05.log
-rw-r--r-- 1 root root 46K Apr 13 06:06 nginx1_2026-04-13_06-06.log
-rw-r--r-- 1 root root 46K Apr 13 06:07 nginx1_2026-04-13_06-07.log
-rw-r--r-- 1 root root 46K Apr 13 06:08 nginx1_2026-04-13_06-08.log
-rw-r--r-- 1 root root 46K Apr 13 06:09 nginx1_2026-04-13_06-09.log
-rw-r--r-- 1 root root 46K Apr 13 06:10 nginx1_2026-04-13_06-10.log
-rw-r--r-- 1 root root 46K Apr 13 06:11 nginx1_2026-04-13_06-11.log
-rw-r--r-- 1 root root 46K Apr 13 06:12 nginx1_2026-04-13_06-12.log
-rw-r--r-- 1 root root 46K Apr 13 06:13 nginx1_2026-04-13_06-13.log
-rw-r--r-- 1 root root 46K Apr 13 06:01 nginx2_2026-04-13_06-01.log
-rw-r--r-- 1 root root 46K Apr 13 06:02 nginx2_2026-04-13_06-02.log
```

Connexion à l'interface web de pfSense :

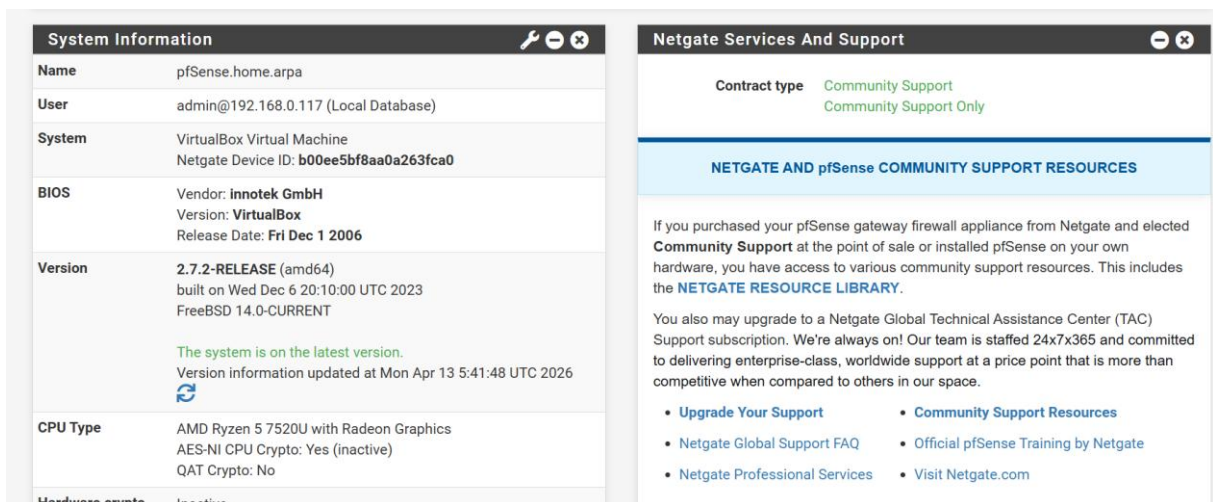


Blocage du trafic non sollicité/suspect :



The screenshot shows the pfSense web interface at the URL `http://192.168.1.1/status_logs_filter.php`. The 'Normal View' tab is selected. The table displays the 'Last 500 Firewall Log Entries. (Maximum 500)'. All entries show a blocked action (red 'X') on the WAN interface, triggered by the rule 'Block private networks from WAN block 192.168/16 (12004)'. The source IP addresses are various private IP ranges, and the destination is 192.168.0.255:15600 using the UDP protocol.

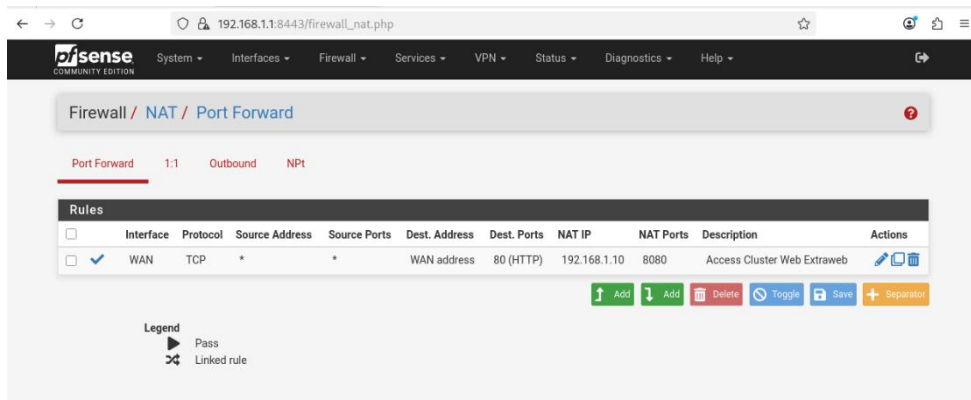
Action	Time	Interface	Rule	Source	Destination	Protocol
×	Apr 12 21:35:39	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:39284	192.168.0.255:15600	UDP
×	Apr 12 21:35:45	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:46081	192.168.0.255:15600	UDP
×	Apr 12 21:35:51	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:33211	192.168.0.255:15600	UDP
×	Apr 12 21:35:57	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:55319	192.168.0.255:15600	UDP
×	Apr 12 21:36:03	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:52846	192.168.0.255:15600	UDP
×	Apr 12 21:36:09	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:32895	192.168.0.255:15600	UDP
×	Apr 12 21:36:15	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:55169	192.168.0.255:15600	UDP
×	Apr 12 21:36:21	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:56710	192.168.0.255:15600	UDP
×	Apr 12 21:36:27	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:56593	192.168.0.255:15600	UDP
×	Apr 12 21:36:33	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:43642	192.168.0.255:15600	UDP
×	Apr 12 21:36:39	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:40479	192.168.0.255:15600	UDP
×	Apr 12 21:36:45	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:40054	192.168.0.255:15600	UDP
×	Apr 12 21:36:51	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:43637	192.168.0.255:15600	UDP
×	Apr 12 21:36:57	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:51949	192.168.0.255:15600	UDP
×	Apr 12 21:37:03	WAN	Block private networks from WAN block 192.168/16 (12004)	192.168.0.142:38899	192.168.0.255:15600	UDP



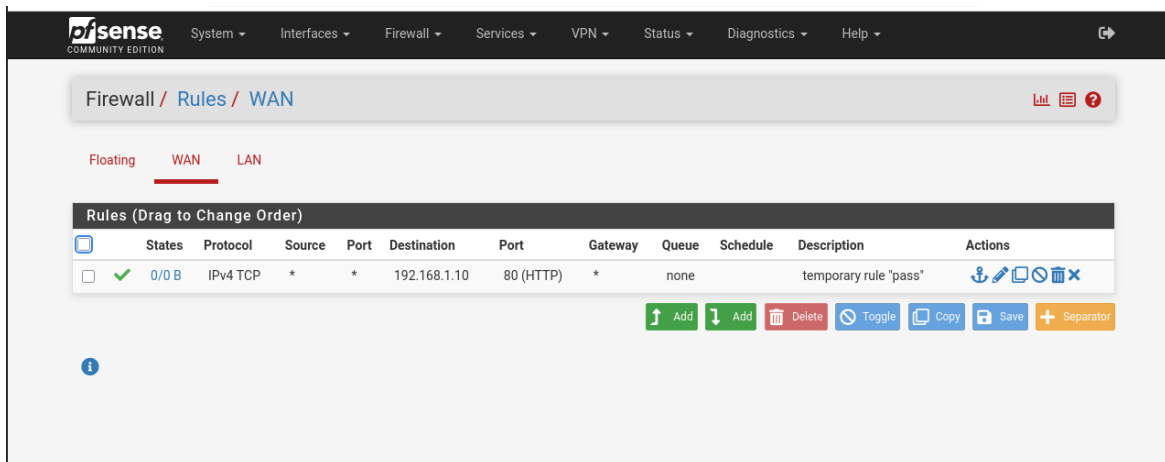
The screenshot shows two side-by-side panels in the pfSense interface. The left panel, 'System Information', provides details about the system's configuration, including the user 'admin@192.168.0.117', the system being a 'VirtualBox Virtual Machine', and the pfSense version '2.7.2-RELEASE (amd64)'. The right panel, 'Netgate Services And Support', shows the 'Contract type' as 'Community Support' and provides links to various support resources.

System Information		Netgate Services And Support	
Name	pfSense.home.arpa	Contract type	Community Support Community Support Only
User	admin@192.168.0.117 (Local Database)		
System	VirtualBox Virtual Machine Netgate Device ID: b00ee5bf8aa0a263fca0		
BIOS	Vendor: innotek GmbH Version: VirtualBox Release Date: Fri Dec 1 2006		
Version	2.7.2-RELEASE (amd64) built on Wed Dec 6 20:10:00 UTC 2023 FreeBSD 14.0-CURRENT The system is on the latest version. Version information updated at Mon Apr 13 5:41:48 UTC 2026		
CPU Type	AMD Ryzen 5 7520U with Radeon Graphics AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No		
Hardware crypto	Inactive		
		NETGATE AND pfSense COMMUNITY SUPPORT RESOURCES	
		If you purchased your pfSense gateway firewall appliance from Netgate and elected Community Support at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the NETGATE RESOURCE LIBRARY. You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.	
		- Upgrade Your Support - Community Support Resources - Netgate Global Support FAQ - Official pfSense Training by Netgate - Netgate Professional Services - Visit Netgate.com	

Règle NAT établie :



Règle ACL :



Connexion à pfSense et validation de « l'accès par pont » sur le même segment :

```
C:\Users\Saema>ping 192.168.0.186
```

```
Envoi d'une requête 'Ping' 192.168.0.186 avec 32 octets de données :
```

```
Réponse de 192.168.0.186 : octets=32 temps=1 ms TTL=64
```

```
Réponse de 192.168.0.186 : octets=32 temps=1 ms TTL=64
```

```
Réponse de 192.168.0.186 : octets=32 temps=1 ms TTL=64
```

```
Réponse de 192.168.0.186 : octets=32 temps=1 ms TTL=64
```

```
Statistiques Ping pour 192.168.0.186:
```

```
Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
```

```
Durée approximative des boucles en millisecondes :
```

```
Minimum = 1ms, Maximum = 1ms, Moyenne = 1ms
```